JEE Mathematics: Quick Review with Grading and Practice Map - Tangent & Normal (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (9 questions; a question that appears in two of these is counted once): High = 2 or more questions, Medium = 1 to 1 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction about tangent and normal

Question 01

Define tangent and normal

Model Answer

Tangent at P: the limiting position of the secant PQ as $Q \to P$ from either side; its slope is $f'(x_1)$.

Normal at P: the line through P perpendicular to the tangent at P.

A tangent may cut the curve again, even at the point of contact: the x-axis is the tangent to $y = x^3$ at O and crosses it there.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

How to find equation of tangent and normal at a point for a curve?

Model Answer

$$T:\ y - y_1 = m(x - x_1), \qquad N:\ y - y_1 = -\frac{1}{m}(x - x_1), \qquad m = \left.\frac{dy}{dx}\right|_{(x_1, y_1)}$$

If $m = 0$: tangent $y = y_1$, normal $x = x_1$. If $m \to \infty$: tangent $x = x_1$, normal $y = y_1$. For an implicit curve differentiate implicitly.

Ex: $y = x^2 + x + 1$ at $(1, 3)$: $m = 3$, $T:\ y = 3x$, $N:\ x + 3y = 10$.

Grade - High

Practice map, main concept: Sheet: J-I 1-2, 4-7, 10, 14; J-II 1-2, 6, 8-9, 58, 80-82; J-M 1, 3; J-A 1.01; C-I 2-3, 24; C-II 4, 6, 15, 51 | DPP: 1(2-4, 6-11, 13-14); 2(8, 14); 6(15) | Question Bank: 150

Also used in: Sheet: J-II 7, 12; J-M 14; C-II 42-44 | DPP: 1(5); 2(11)

Question 03

How to find equation of tangent and normal at a point for a curve whose equation is given in parametric form?

Model Answer

$$m = \frac{dy}{dx} = \left.\frac{dy/dt}{dx/dt}\right|_{t=t_0}, \qquad (x_1, y_1) = (x(t_0),\ y(t_0))$$

Then use the point-slope forms of item 2.

Ex: $x = 2t,\ y = t^2$ at $t = 2$: point $(4, 4)$, $m = t = 2$; $T:\ 2x - y = 4$, $N:\ x + 2y = 12$.

Grade - Low

Practice map, main concept: Sheet: J-I 3; J-II 5, 111; C-I 25; C-II 3

Also used in: Sheet: J-II 3-4; C-I 5; C-II 16, 51 | DPP: 2(3)

Question 04

How to find equation of tangent and normal at a point for a curve which is a non-uniform function?

Model Answer

At a junction point check continuity and then LHD and RHD. Equal finite LHD = RHD gives the slope; LHD and RHD unequal (corner) means no unique tangent there.

Away from the junction, use the formula of the piece that holds there (open | | first).

Ex: $f = x - 2$ $(x \leq 1)$, $x^2 - x - 1$ $(x > 1)$: $f(1) = -1$, $f'(1^-) = f'(1^+) = 1$; $T:\ x - y - 2 = 0$, $N:\ x + y = 0$.

Grade - **Low**

Practice map, main concept: Sheet: J-II 4

Also used in: Sheet: C-I 23

Objective Concepts

Question 05

What will be slope of tangent at x = 0 for an even function?

Model Answer

f even $\Rightarrow$ f' odd $\Rightarrow$ $f'(0) = 0$ (if it exists): the tangent at $x = 0$ is horizontal, $y = f(0)$.

Ex: $y = \cos x + x^2$: tangent at $x = 0$ is $y = 1$.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 06

How to write equation of tangent at origin for a polynomial curve passing through origin?

Model Answer

Equate the lowest-degree terms in x and y to zero.

Ex: $x^2 + y^2 - 4x + 3y = 0 \Rightarrow T:\ 3y - 4x = 0$; $x^3 + y^3 - 3xy = 0 \Rightarrow xy = 0$ (two tangents $x = 0,\ y = 0$).

Grade - **Low**

Practice map, main concept: Sheet: J-I 13; C-I 20

Also used in: none

Vertical Tangent

Question 07

How to check whether vertical tangent exists at a point for a function?

Model Answer

f continuous at x_0 and LHD, RHD both $+\infty$ or both $-\infty$ $\Rightarrow$ vertical tangent $x = x_0$. One $+\infty$ and one $-\infty$ (cusp) $\Rightarrow$ no vertical tangent.

For an implicit curve: vertical tangent where $\frac{dx}{dy} = 0$.

Ex: $y = x^{1/3}$ has the vertical tangent $x = 0$; $y = x^{2/3}$ and $y = |x|$ have none at 0. $y^3 + 3x^2 = 12y$: vertical tangents at $\left(\pm\frac{4}{\sqrt{3}}, 2\right)$.

Grade - **Low**

Practice map, main concept: Sheet: C-I 23 | DPP: 1(1, 12)

Also used in: Sheet: C-I 3

Some important parametric coordinates

Question 08

Write parametric coordinate for standard second degree curves.

Model Answer

$$x^2+y^2=a^2:\ (a\cos\theta,\ a\sin\theta);\quad \frac{x^2}{a^2}+\frac{y^2}{b^2}=1:\ (a\cos\theta,\ b\sin\theta);\quad \frac{x^2}{a^2}-\frac{y^2}{b^2}=1:\ (a\sec\theta,\ b\tan\theta)$$

$$xy=c^2:\ \left(ct,\ \frac{c}{t}\right);\quad y^2=4ax:\ (at^2,\ 2at);\quad x^2=4ay:\ (2at,\ at^2)$$

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 2 | DPP: 2(8) | Question Bank: 152

Question 09

Write parametric coordinate for $\dfrac{x^n}{a^n}+\dfrac{y^n}{b^n}=1$.

Model Answer

$$\left(a\cos^{2/n}\theta,\ b\sin^{2/n}\theta\right)$$

Ex: astroid $x^{2/3}+y^{2/3}=a^{2/3}$ $(n=\frac{2}{3})$: $(a\cos^3\theta,\ a\sin^3\theta)$, slope $-\tan\theta$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-II 2 | DPP: 1(7, 10)

Question 10

How to assume point on a curve whose equation is known?

Model Answer

$y=f(x)$: take $(t,\ f(t))$; line $y=mx+c$: $(t,\ mt+c)$; standard conics: the parametric forms of item 8. One parameter keeps the work to one unknown.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 1, 7-8; J-II 10-11; C-I 24; C-II 1, 3-4, 6, 45 | DPP: 1(6); 6(15)

Ultimate Concept

Question 11

What is ultimate concept?

Model Answer

Always equate the slope of the tangent at the unknown point $P(t, f(t))$ with the slope of the line through P and the other known point.

$$f'(t)=\frac{f(t)-y_1}{t-x_1}$$

Ex: tangent to $y=x^3$ at $P(2,8)$ meets it again at $Q(t,t^3)$: $\frac{t^3-8}{t-2}=12\Rightarrow t^2+2t-8=0\Rightarrow Q(-4,-64)$.

Grade - Low

Practice map, main concept: Sheet: J-I 8; J-II 10; C-II 1, 16

Also used in: Sheet: J-II 9

Question 12

How to find equation of a tangent/normal passing through a point?

Model Answer

Tangent through $A(x_1, y_1)$: $f'(t) = \frac{f(t)-y_1}{t-x_1}$. Normal through A: $-\frac{1}{f'(t)} = \frac{f(t)-y_1}{t-x_1}$. Solve for t; each root gives one line.

Ex: normal to $x^2 = 4y$ through $(1, 2)$: $P(2t, t^2)$, $-\frac{1}{t} = \frac{t^2-2}{2t-1} \Rightarrow t^3 = 1 \Rightarrow P(2, 1)$, $N: \; x + y = 3$.

Grade - Low

Practice map, main concept: Sheet: J-II 3, 11; C-II 40, 45 | DPP: 2(10)

Also used in: Sheet: J-I 9

Question 13

How to find equation of common tangent/normal to two curves?

Model Answer

$$P(t_1, f(t_1)),\; Q(t_2, g(t_2)): \quad f'(t_1) = g'(t_2) = \frac{g(t_2) - f(t_1)}{t_2 - t_1}$$

For a common normal use $-\frac{1}{f'(t_1)} = -\frac{1}{g'(t_2)} =$ slope of PQ.

Ex: $y = x^2 + 1$, $y = -x^2$: $2t_1 = -2t_2$, $t_1^2 = \frac{1}{2}$; common tangents $y = \pm\sqrt{2}\,x + \frac{1}{2}$.

Grade - Low

Practice map, main concept: Sheet: J-II 108

Also used in: none

Minimum Distance between two curves

Question 14

How to find minimum/maximum distance of a point from a curve?

Model Answer

AP (A fixed, P on the curve) is extreme when AP is the normal at P: (slope of AP) $\times f'(P) = -1$. For a circle centre C: distance to the curve $= CP - r$.

Ex: $y = x^2 + 7$ and $A(3, 7)$: $2t \cdot \frac{t^2}{t-3} = -1 \Rightarrow t = 1$, $P(1, 8)$, $d_{\min} = \sqrt{5}$.

Grade - High

Practice map, main concept: Sheet: J-I 9, 19; J-II 13, 110; J-M 2, 35-36 | Question Bank: 149, 151-152

Also used in: Sheet: J-I 11

Question 15

What does minimum distance between two curves represent?

Model Answer

Zero if the curves meet; otherwise it is measured along their common normal (the shortest segment is normal to both curves).

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-II 14 | Question Bank: 147

Question 16
How to find minimum/maximum distance between two curves?

Model Answer
Line and curve: find P on the curve where the tangent is parallel to the line; answer = distance of P from the line. Circle and curve: normal at P through the centre; answer $= CP - r$. Otherwise find the common normal.
Ex: $y = x^2 + 3x + 2$ and $y = x - 2$: $2t + 3 = 1 \Rightarrow P(-1, 0)$, $d = \frac{|-1-0-2|}{\sqrt{2}} = \frac{3}{\sqrt{2}}$. $x^2 + y^2 = 2$ and $xy = 9$: $P(3, 3)$, $d = 3\sqrt{2} - \sqrt{2} = 2\sqrt{2}$.

Grade - High
Practice map, main concept: Sheet: J-I 11-12; J-II 14-15 | DPP: 2(13) | Question Bank: 147
Also used in: Sheet: J-I 19; J-M 35 | Question Bank: 149

Angle of Intersection between two curves

Question 17
What is angle of intersection between two curves?

Model Answer
The acute angle between the tangents to the two curves at their common point.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 18
What will be angle of intersection between two curves if they do not have any common point?

Model Answer
It is not defined: first check that the curves really meet (solve them together).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 19
How to find angle of intersection between two curves?

Model Answer

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

m_1, m_2 = slopes of the two curves at the common point. $m_1 m_2 = -1 \Rightarrow \theta = 90°$; one slope infinite $\Rightarrow \tan\theta = \left|\frac{1}{m}\right|$.
Ex: $y = x^2$ and $6y = 7 - x^3$ at $(1, 1)$: $m_1 = 2$, $m_2 = -\frac{1}{2}$, $\theta = \frac{\pi}{2}$.

Grade - Low
Practice map, main concept: Sheet: J-II 16; C-II 5 | DPP: 2(1, 5, 11)
Also used in: Sheet: J-I 13; J-II 17; C-II 51

Question 20
What will be angle of intersection if two curves have more than one common point?

Model Answer
Find the angle at each common point separately; the angles can differ.
Ex: $2y^2 = x^3$, $y^2 = 32x$: $90°$ at $(0,0)$; $\tan^{-1}\frac{1}{2}$ at $(8, \pm 16)$ ($m_1 = 3$, $m_2 = 1$).

Grade - Low
Practice map, main concept: Sheet: J-II 17
Also used in: none

Orthogonal curves

Question 21
What is the condition for two curves to be orthogonal?

Model Answer
Angle of intersection $90°$ at every common point: $m_1 m_2 = -1$ there (or one tangent horizontal and the other vertical).
Ex: $y^2 = 4ax$ and $y = e^{-x/2a}$: $m_1 = \frac{2a}{y_1}$, $m_2 = -\frac{y_1}{2a}$, product -1: orthogonal.

Grade - Low
Practice map, main concept: Sheet: J-I 15-16; J-II 19
Also used in: Sheet: J-II 18

Question 22
How to check orthogonality of two curves without finding common point?

Model Answer
Write $m_1 m_2$ at a general common point (x_1, y_1) and use both curve equations to show it equals -1 (or to get the condition).

$$a_1x^2 + b_1y^2 = 1 \ \perp \ a_2x^2 + b_2y^2 = 1 \iff \frac{1}{a_1} - \frac{1}{b_1} = \frac{1}{a_2} - \frac{1}{b_2}$$

Ex: $xy = a^2$ and $x^2 - y^2 = b^2$: $m_1 = -\frac{y_1}{x_1}$, $m_2 = \frac{x_1}{y_1}$, $m_1 m_2 = -1$ always.

Grade - Low
Practice map, main concept: Sheet: J-II 18
Also used in: none

Case when two curves touch each other

Question 23
How to check whether two curves touch each other?

Model Answer

$$f(x_1) = g(x_1), \quad f'(x_1) = g'(x_1)$$

Common point with equal slopes. "Touches the x-axis at $x = a$" means $f(a) = 0$ and $f'(a) = 0$.
Ex: $x^2 + y^2 = 8$ and $xy = 4$ at $(2,2)$: both slopes -1, so they touch.

Grade - Medium
Practice map, main concept: Sheet: J-II 7, 109; J-M 14; C-I 5; C-II 2 | DPP: 1(5)
Also used in: Sheet: J-I 17-18; J-II 86-87; C-I 4

Question 24

How to analyse equation $e^x = kx$, $\ln x = kx$ and $\ln x = kx^2$ for number of solutions?

Model Answer

Find the touching value of k (common point and equal slopes), then rotate the line / curve about it.

$e^x = kx$: touch at $k = e$ (point $(1, e)$). $k < 0$: one; $0 \le k < e$: none; $k = e$: one; $k > e$: two.

$\ln x = kx$: touch at $k = \frac{1}{e}$ $(x = e)$. $k \le 0$: one; $0 < k < \frac{1}{e}$: two; $k = \frac{1}{e}$: one; $k > \frac{1}{e}$: none.

$\ln x = kx^2$: touch at $k = \frac{1}{2e}$ $(x = \sqrt{e})$. $k \le 0$: one; $0 < k < \frac{1}{2e}$: two; $k = \frac{1}{2e}$: one; $k > \frac{1}{2e}$: none.

Grade - Low

Practice map, main concept: Sheet: J-I 17-18; J-II 86-88; C-I 4; C-II 46

Also used in: none

Some Important Lengths

Question 25

Define length of tangent, normal, subtangent and subnormal for a point on a curve

Model Answer

Tangent and normal at P meet the x-axis at T and G; M = foot of the ordinate of P. Length of tangent = PT, of normal = PG, subtangent = TM, subnormal = MG.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 26

Write results for length of tangent, normal, subtangent and subnormal

Model Answer

$$PT = \left|\frac{y_1\sqrt{1+m^2}}{m}\right|, \quad PG = \left|y_1\sqrt{1+m^2}\right|, \quad TM = \left|\frac{y_1}{m}\right|, \quad MG = |y_1 m|$$

$m = \frac{dy}{dx}$ at $P(x_1, y_1)$.

Ex: $y^2 = 8x$ at $(2, 4)$: $m = 1$; $PT = 4\sqrt{2}$, $TM = 4$, $PG = 4\sqrt{2}$, $MG = 4$.

Grade - Low

Practice map, main concept: Sheet: J-I 20; J-II 12, 20-21 | DPP: 2(2-3, 12)

Also used in: Sheet: J-II 111

Question 27

Prove results for length of tangent, normal, subtangent and subnormal by both methods

Model Answer

Method 1: $\tan\psi = m$, $\angle MPG = \psi$: $PT = y\,\mathrm{cosec}\,\psi$, $TM = y\cot\psi$, $PG = y\sec\psi$, $MG = y\tan\psi$.

Method 2: put $y = 0$ in T and N: $T\left(x_1 - \frac{y_1}{m}, 0\right)$, $G(x_1 + my_1, 0)$; then the distance formula.

Grade - Low

Practice map, main concept: none

Also used in: none

Approximation

Question 28

How to approximate any square root, cube root etc of a number with the help of differentials

Model Answer

$$f(x+\Delta x) \approx f(x) + f'(x)\,\Delta x, \qquad dy = f'(x)\,dx \approx \Delta y$$

Choose x near the number with $f(x)$ known exactly, and small Δx.
Ex: $\sqrt{36.6} \approx 6 + \frac{0.6}{2\cdot 6} = 6.05$; $\sqrt{25.2} \approx 5 + \frac{0.2}{10} = 5.02$.

Grade - Low

Practice map, main concept: Sheet: C-II 13

Also used in: none

Question 29

What is the geometrical meaning of approximation with the help of differentials?

Model Answer

Near $P(x, y)$ the curve is replaced by its tangent: dy is the rise along the tangent, Δy the rise along the curve; for small $dx = \Delta x$, $dy \approx \Delta y$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 30

How to check whether approximate value is greater/less than actual value?

Model Answer

$f'' > 0$ (concave up): tangent below the curve, approximate value $<$ actual. $f'' < 0$ (concave down): tangent above, approximate value $>$ actual.
Ex: $\sqrt{x}$ has $f'' < 0$, so $6.05 > \sqrt{36.6} = 6.0498\ldots$

Grade - Low

Practice map, main concept: none

Also used in: none

Miscellaneous

Question 31

How to find the rate of change of a quantity, and solve related-rate problems (rate measure)?

Model Answer

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

Write one relation between the quantities valid at every instant, differentiate w.r.t. t, then put the values of that instant. Increasing rate > 0, decreasing rate < 0.

Cone of water / sand: use similar triangles to remove r or h before differentiating.

Ex: sand cone with $h = \frac{r}{6}$: $V = 12\pi h^3$; $\frac{dV}{dt} = 12$ at $h = 4$ gives $\frac{dh}{dt} = \frac{12}{36\pi\cdot 16} = \frac{1}{48\pi}$ cm/s.

Grade - Medium

Practice map, main concept: Sheet: J-I 21, 23; J-II 22, 24-27; C-I 6; C-II 7-12, 14, 42-44 | DPP: 2(4, 6-7, 9) | Question Bank: 148

Also used in: none

Question 32

For a point moving on a curve, how to compare the rates of change of its abscissa and ordinate?

Model Answer

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

Ordinate changes k times as fast as the abscissa: $\frac{dy}{dx} = k$. Abscissa faster than ordinate: $\left|\frac{dy}{dx}\right| < 1$; slower: $\left|\frac{dy}{dx}\right| > 1$.

Ex: $6y = x^3 + 2$, y changes 8 times as fast as x: $\frac{x^2}{2} = 8 \Rightarrow (4, 11)$ and $\left(-4, -\frac{31}{3}\right)$.

Grade - **Low**

Practice map, main concept: Sheet: J-I 22; J-II 23

Also used in: none